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TECHNICAL SUMMARY

Transmission Line Cost Tool

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ABSTRACT

An automated Excel/VBA tool that estimates transmission line cost from voltage level, length and terrain. An automated cost-estimation tool for transmission line projects, driven by voltage level, line length and terrain.

1. Problem and motivation

Early in planning, engineers need a quick, consistent cost estimate for alternative line routes and voltage levels, long before a detailed design exists.

2. Methodology

- Designed calculation algorithms for material, installation and infrastructure costs
- Automated the estimate in Excel/VBA
- Enabled fast scenario analysis across design configurations

3. Figures

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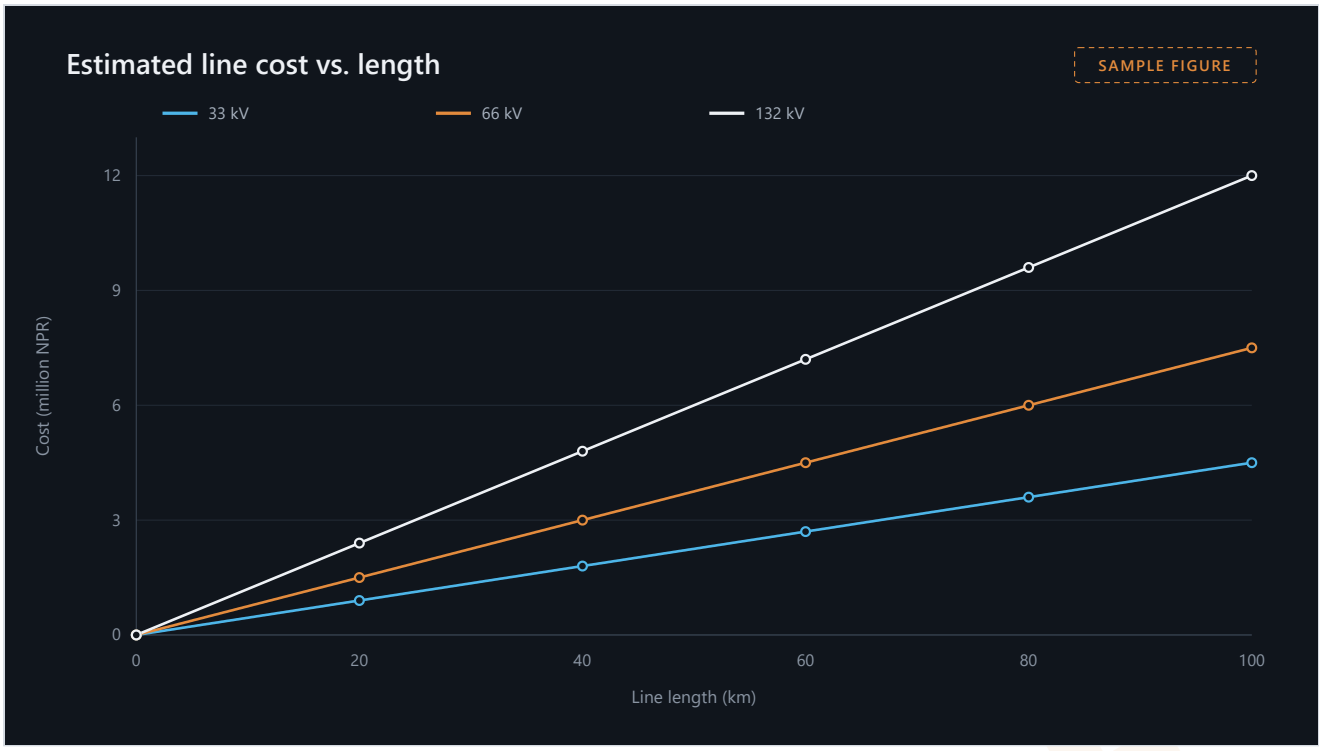


Figure 1. Estimated line cost against length for three voltage levels.

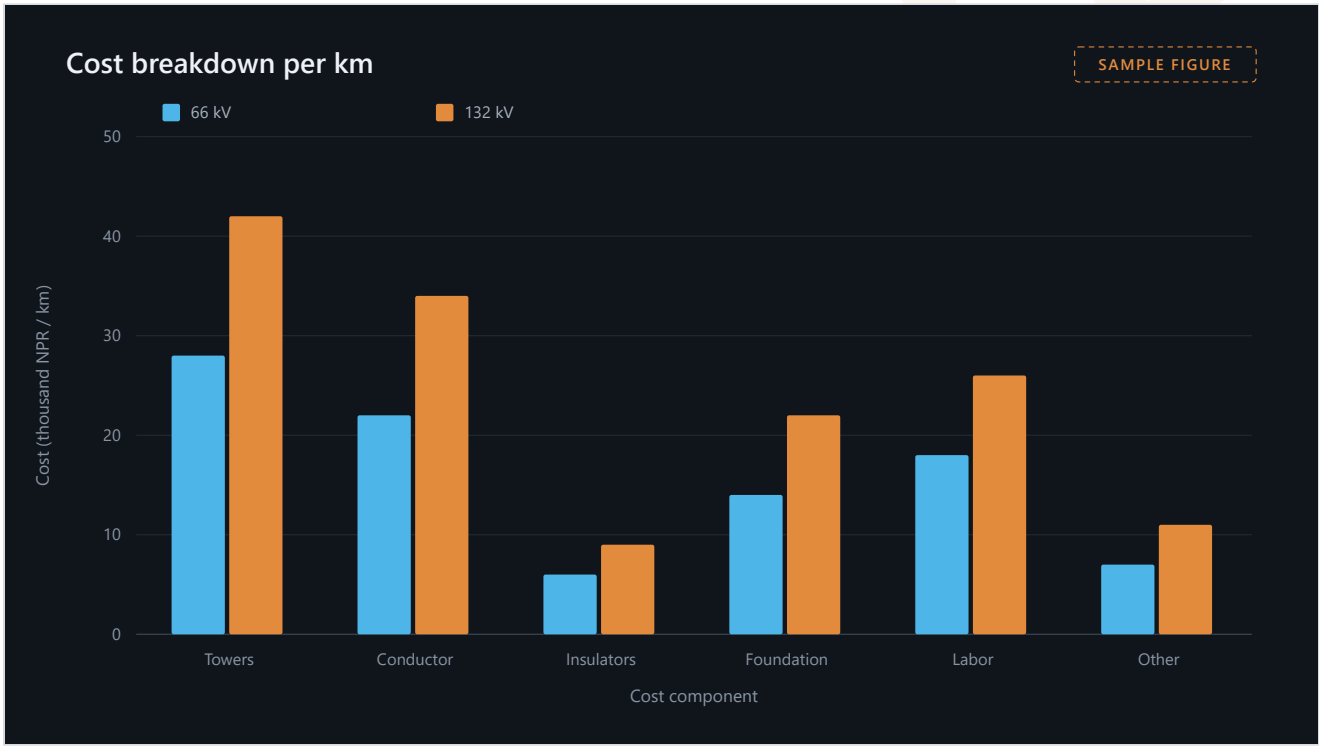


Figure 2. Cost breakdown per km by component.

4. Results

- Supports planning and budgeting decisions with faster, more consistent estimates

5. Team and contribution

Contribution: Designed the calculation algorithms and programmed the tool.

Tools: Excel/VBA, C++

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